Semester II, Academic Session 2025/2026

CKS121/CSE242 Software Requirements

Assignment I and Assignment II Instructions

I. Overview

The purpose of this assignment is to expose students to software requirements activities such as requirement elicitation, documentation, and management. This document consists of assignment I and assignment II instructions.

1. Form a group that consists of **FIVE (5) members**, select ONE (1) leader, and post your group information together with the project title on the link provided in e-learn.

2. Identify or select any software development project (either new or improvement of existing software/system) that you want to work on. Please check that your group project title is NOT SIMILAR to other group's before you upload the title in the form.

3. Identify requirements sources and elicitation techniques.
    a. Each member needs to select **ONE (1) elicitation technique** and requirement gathering sessions (for example, focus group, interview, surveys, observation, walkthrough, brainstorming – any suitable techniques that involve stakeholders/ potential users/end users).
    b. For each technique selected in (a), each member is required to explain about the technique setting, the advantages and disadvantages of the technique in relation with your proposed system's requirement elicitation.
    c. Discuss, select and **CONDUCT** elicitation and requirement gathering sessions based on your assignment findings in (a) and (b), and you may include other supporting elicitation techniques such as reviewing existing documents, reviewing competitor's systems etc. to support your requirements.
    d. Prepare and deliver a 10-minute pitch based on items (a), (b) and (c). You **MUST** conduct any elicitation activities and present your findings with suitable number of respondents *(ie. less than 5 respondents are not allowed for Interview, acceptable number for survey respondents are within 30-50 respondents).* DO NOT ANSWER YOUR OWN INSTRUMENT OR FABRICATE DATA.
    e. Group leader must submit the video recording in e-learn by **Week 9**. **Assignment I contributes 20% of your coursework grade.**

4. Submit a report that includes the following sections as listed in **II. Specification report** by **Week 14** via e-learn. **Assignment II contributes 10% of your coursework grade.**

II. Specification report

Table of contents (Use as a guideline)

1. Project description

<Provide a concise summary of the motivation for the project. This section could describe the business problem the project is intended to solve including the system goal.

Situation *Describe the background, context, and environment.*
Problem *Describe the business problem (or problems) as you now understand it*
Implication *Describe the likely results if the problem isn't solved.*
Benefit *State the business value of solving the problem.*
Vision *Describe what the desired future state would look like.>*

2. List of Stakeholders

<Stakeholders are individuals, groups, or organizations that are actively involved in a project, are affected by its outcome, or can influence its outcome. The stakeholder profiles identify the project sponsor, project manager, customers for this product, and other stakeholders.>

3. Visions

<Summarizes the purpose and intent of the new product/system. This is a good template for a vision statement:

For *target customer*
Who *statement of the need or opportunity*
The *product name*
Is *product category*
That *major capabilities, key benefit, compelling reason to buy or use*
Unlike *primary competitive alternative, current system, current business process (if any)*
Our product *statement of primary differentiation and advantages of new product>*

4. Elicitation techniques

<Select relevant elicitation techniques, conduct the requirement gathering sessions and report your findings. Explain how the requirement elicitation facilitate requirements gathering of your proposed system> This is based on Assignment I.

5. System Requirements Specification

<Incorporate this template for this section. R1, R2… etc. stands for requirement 1, requirement 2..etc. Type includes functional and non-functional requirements. Example of an online shopping platform (JomBeli) as follows>

ID	Details	Type	Priority
R1	JomBeli shall display a list of all products offered by all registered sellers in the platform.	• Products • Functional	MustHave
R2	JomBeli shall automatically calculate and add a shipping charge to the order made.	• Payment • Functional	ShouldHave
R3	JomBeli shall allow a customer to cancel order provided that the order has not been dispatched.	• Order • Functional	MustHave
R4	JomBeli shall be available 24 hours per day, 360 days per year.	• Availability • Non-Functional	MustHave
R5	JomBeli shall log in a customer within 5 seconds.	• Performance • Non-Functional	ShouldHave
R6	JomBeli shall use secure means of communication.	• Security • Non-Functional	MustHave
	…	•	…

	…	•	…
	…	•	…

6. Use Cases

6.1 Use case diagram

<Include use case diagram>

6.2 Actor semantics

<Incorporate this template for this section. Example of an online shopping platform (JomBeli) as follows>

Actor	Description
Customer	Someone who buys products from JomBeli platform
SystemAdministrator	A user of the system who can set up access rights for other users
CardProcessingCompany	An external company that processes credit card transactions on behalf of JomBeli platform
…	…
…	…

6.3 Use case Descriptions

<Incorporate this template for this section. Use case descriptions are required for each use case in the use case diagram. Example of an online shopping platform (JomBeli) as follows>

Use case:	*<ID and use case name>* Example: Use case ID 1: BrowseProducts
Actor(s):	Customer
Description:	*<Provide a brief description of the reason for and outcome of this use case>* Example: Customer can browse products from the catalogue and display the detail descriptions of the product including picture, product descriptions, product review and rating and seller information.
Preconditions:	*<List any activities that must take place, or any conditions that must be true, before the use case can be started>* Example: None
Flow of events:	*<Provide a description of the user actions and corresponding system responses that will take place during execution of the use case under normal, expected conditions>* Example: 1. Customer search for particular product 2. The system displays a list of products relevant to the search product 3. The customer select one product to see detail description of the product 4. The system displays the product details
Postconditions:	*<Describe the state of the system at the successful conclusion of the use case execution.>* Example: None
Other Information (optional):	*<Identify any additional pertinent requirements or constraints for the use case, such as quality attributes. Describe what should happen if the use case execution fails for some unanticipated or systemic reason (e.g., loss of network access, timeout)>*

Use case:	*<ID and use case name>* Example: Use case ID 2: AddToCart
Actor(s):	Customer
Description:	*<Provide a brief description of the reason for and outcome of this use case>* Example: Customer can add to cart selected products.
Preconditions:	*<List any activities that must take place, or any conditions that must be true, before the use case can be started>* Example: 1. The Customer is browsing products.

Flow of events:	*<Provide a description of the user actions and corresponding system responses that will take place during execution of the use case under normal, expected conditions>* Example: 1. The customer selects a product 2. The customer selects “Add item” 3. The system adds the items to the customer’s shopping basket 4. Include(DislayBasket) *if you have include in the use case diagram
Postconditions:	*<Describe the state of the system at the successful conclusion of the use case execution.>* Example: 1. A product has been added to the customer’s basket 2. The contents of the basket are displayed.
Other Information (optional):	*<Identify any additional pertinent requirements or constraints for the use case, such as quality attributes. Describe what should happen if the use case execution fails for some unanticipated or systemic reason (e.g., loss of network access, timeout)>*

7. **Appendix**
   <Attach your questionnaire/interview questions/meeting minutes/documents review etc. in this section> This is based on Assignment I.

8. **Conclusion and reflection** (what you learnt from these assignments and discuss how these assignments help you to finalize the system requirements).

Important:

Your assignments must be your own group work. You will be assigned 0 marks for this entire assignment, if any of your group answers bears a close resemblance to another group’s submission, or to something previously published on the Internet or elsewhere.

Any questions regarding these assignments, please email us at athiyah@usm.my or

nursakirah.arm@usm.my (for CKS121W/CSE242W) .Thank you.